

Project Overview — DIU FoundHub

A Secure Lost and Found Platform for Daffodil International University

1 Introduction

Students and staff frequently lose personal belongings such as student identity cards, mobile phones, books, bags, keys, documents, and accessories on campus. Without a single trusted system, reports are scattered across social-media posts, informal group messages, or notice boards. This makes it difficult to find the correct owner, preserve privacy, and maintain an accountable record of the handover.

DIU FoundHub is a browser-based Lost and Found platform designed for Daffodil International University. It provides one secure place to report lost or found items, attach photographs, search active listings, submit ownership claims, and complete staff-verified returns. The system connects students, staff, and administrators through a clear workflow instead of relying on informal communication.

The project is not only a listing board. It is a structured campus recovery system that combines manual account registration, protected item data, ownership verification, status tracking, and event-driven notifications.

2 Problem Statement

The university community needs a more reliable way to manage lost belongings. Existing informal approaches are difficult to search, can expose private information, and do not provide a verifiable record of who claimed or received an item. A finder may not know where to publish a report, while an owner may not know whether an item has already been handed to a department or security desk.

Typical challenges include:

- Lost and found information is dispersed across multiple unofficial channels.
- Item photos, locations, and dates are often incomplete or difficult to filter.
- Anyone can claim an item without providing sufficient proof of ownership.
- There is no consistent visibility into whether an item is lost, claimed, verified, or returned.
- Staff need a secure way to review claims and document a physical handover.

DIU FoundHub addresses these problems through a database-backed workflow that preserves report history, restricts private claim evidence, and gives authorized staff a dedicated verification process.

3 Vision

To build a user-friendly digital campus service that makes returning belongings safer, faster, and more transparent. DIU FoundHub aims to provide a dependable recovery network where every report can be discovered, every ownership claim can be reviewed, and every completed return can be recorded.

This vision supports a more responsible campus environment by turning lost-and-found reporting into a structured, searchable, and accountable process.

4 Core Capabilities

Module	Implementation	User Benefit
Manual Account Access	Registration with name, student ID, Gmail address, and a securely hashed password; protected browser sessions and logout.	Creates accountable user access without exposing passwords or requiring external sign-in.

Module	Implementation	User Benefit
Lost and Found Reports	Forms for title, description, category, date, location, contact details, holding location, and item photo.	Captures complete, consistent information for faster discovery and response.
Public Discovery	Search, category/date/status filters, sorting, pagination, item detail pages, and responsive browsing.	Helps students and staff locate relevant reports quickly on desktop or mobile browsers.
Ownership Claims	Private proof description and unique identifiers; self-claim and duplicate-claim prevention.	Protects belongings by requiring meaningful evidence before an item is released.
Staff Verification	Administrator dashboard for reviewing claims, approving or rejecting decisions, managing listings, and confirming returns.	Gives DIU staff an auditable role in the final handover process.
Notifications	In-app alerts for possible matches, new claims, and claim decisions; optional email delivery audit.	Keeps relevant users informed at the time an important event occurs.

5 Innovation and Differentiation

Unlike a static notice board, DIU FoundHub combines several practical web technologies into one connected university service:

- A secure manual registration and password-login workflow tailored to student identity information.
- A complete Lost → Claimed → Verified → Returned status lifecycle, visible to the reporter.
- Private ownership evidence that is accessible only to the claimant and authorized staff reviewers.
- Complementary item-match detection between lost and found reports in the same category.
- Cloud-backed item-photo storage with database records that retain only image references.
- Role-based administration for claim decisions, item management, and verified returns.

In short, DIU FoundHub is designed as a realistic, database-driven campus application rather than a static demonstration site.

6 Expected Outcomes

Upon completion and use, the project is expected to demonstrate:

- A fully responsive web platform for reporting and browsing lost and found items.
- Secure user registration, login, profile completion, and session-based access control.
- Dynamic CRUD operations for item reports, ownership claims, notifications, and profile data.
- A clear verification workflow that reduces incorrect item handovers.
- A staff dashboard that supports organized review, decision-making, and return confirmation.
- A maintainable full-stack implementation using React, TypeScript, Express, tRPC, MySQL/TiDB, Drizzle ORM, and cloud-compatible storage.

7 Project Team Roles (For Understanding)

- **Project Lead / Full-Stack Developer** — manages the project plan, user interface, server procedures, testing, and final documentation.

- **Frontend Developer** — designs the responsive pages, forms, navigation, filters, status timeline, and dashboard experience.
- **Backend Developer** — implements authentication, authorization, validation, CRUD operations, claims, matching, and notification events.
- **Database Designer** — designs the relational schema, migrations, indexes, status history, and data-integrity rules.
- **QA and Security Reviewer** — checks manual authentication, ownership restrictions, input validation, test coverage, and responsive behavior.
- **DIU Staff Administrator** — reviews sensitive claims, verifies proof, records decisions, and confirms item handovers.

8 Conclusion

DIU FoundHub transforms a common campus problem into a clear digital service. It connects item reporters, finders, owners, and staff through one secure workflow that supports discovery, verification, and accountable returns.

By combining dynamic database storage, manual password-protected access, role-based administration, private ownership claims, notifications, cloud image support, and an exact return-status lifecycle, the project provides a practical foundation for a modern university Lost and Found system.